Development of Novel AI Models for Early Detection and Prediction of Infectious Diseases, with a Focus on Sepsis

Can a sepsis warning stay useful in a hospital it has never seen?

Mahmoud Essam Elrggal
PhD candidate, School of Computing, Engineering and the Built Environment, Edinburgh Napier University

Director of Studies: Prof. Ahmed Al Dubai
Second Supervisor: Dr Kia Dashtipour



SCEBE Research Degree Conference 2026

20 October 2026
Merchiston Campus
Edinburgh Napier University

CONTEXT

Detecting infection early saves lives and saves antibiotics

11 M

sepsis deaths a year: 1 in 5 deaths worldwide [1]

1.27 M

deaths in 2019 caused directly by antibiotic-resistant bacteria; 4.95 M associated [2]

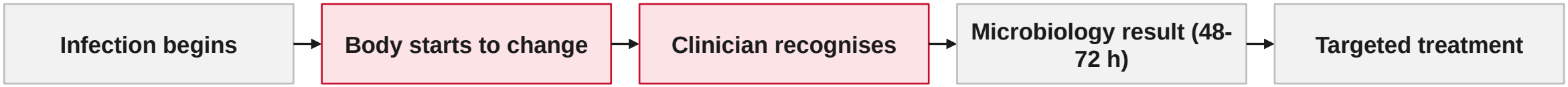
OR 0.66

odds of death with rapid diagnosis of bloodstream infection; effective therapy about 5 h sooner [3]

-18.7 %

relative in-hospital mortality when a sepsis AI alert was confirmed by clinicians within 3 h [4]

The infection pathway



AI prediction moves recognition earlier

AI rapid diagnostics shorten this step [5]

[1] Rudd et al., Lancet 2020 [2] AMR Collaborators, Lancet 2022 [3] Timbrook et al., Clin Infect Dis 2017 [4] Adams et al., Nat Med 2022 [5] Elrggal et al., submitted 2026

Sepsis is an infection that makes the body's own organs fail

Sepsis-3: life-threatening organ dysfunction caused by a dysregulated host response to infection [6]

Organ failure is measured by the SOFA score: 0 to 4 points for each organ system [7]



Lungs

oxygen ratio PaO2/FiO2



Circulation

blood pressure, vasopressors



Kidneys

creatinine, urine output



Liver

bilirubin



Clotting

platelets



Brain

Glasgow Coma Scale

Why it is hard

- Sepsis contributes to between one in two and one in three hospital deaths [8].
- Each hour of delay to antibiotics raises the odds of death by about 4 % [9]; in septic shock survival falls by 7.6 % per hour [10].
- Early signs are non-specific: fever, a fast heart rate and fast breathing are common in many ICU patients.
- Bedside screening scores such as qSOFA and SIRS detect deterioration less accurately than early-warning scores [11].

[6] Singer et al., JAMA 2016 [7] Vincent et al., Intensive Care Med 1996 [8] Liu et al., JAMA 2014 [9] Seymour et al., N Engl J Med 2017 [10] Kumar et al., Crit Care Med 2006 [11] Churpek et al., Am J Respir Crit Care Med 2017

Clinical AI often excels in development and fails in practice

It fails on new data

81 % of 86 deep-learning algorithms performed worse on external data [12]. A model fell from AUC 0.90 to 0.50 when run silently on new patients [13]. Of 81 deep-learning studies, 9 were prospective and 6 tested in real clinical settings [14].

It sees a narrow picture

The most used sepsis benchmark gives models 40 hourly variables [15]; treatments, devices and history are usually left out.

It learns the care process, not the patient

Whether a laboratory test was ordered predicted survival for 86 % of 272 tests; the timing of the order beat the result itself for 68 % [16]. A pneumonia model learned which hospital took the X-ray [17].

It is judged by numbers, not clinical meaning

Data leakage inflates results in at least 17 scientific fields [18]. Usefulness depends on the action a prediction triggers, not on AUROC alone [19, 20].

[12] Yu et al., Radiol Artif Intell 2022 [13] Kwong et al., Front Digit Health 2022 [14] Nagendran et al., BMJ 2020 [15] Reyna et al., Crit Care Med 2020 [16] Agniel et al., BMJ 2018 [17] Zech et al., PLoS Med 2018 [18] Kapoor et al., Patterns 2023 [19] Shah et al., JAMA 2019 [20] Kelly et al., BMC Med 2019

Sepsis AI follows the same pattern

PhysioNet 2019 Challenge: 104 teams [15]

0.43 → **-0.12**

winner's utility on new patients from its two training hospitals

on a third hospital it had never seen: worse than no alarm [21]

Epic Sepsis Model, widely deployed [22]

0.63

AUROC in independent validation (vendor: 0.76 to 0.83)

67 %

of sepsis cases missed while alerting on 18 % of all admissions

Across the field: 28 sepsis models reported AUROC from 0.68 to 0.99, with almost no prospective testing [23]. Studies use different definitions and implementations of sepsis, so results cannot be compared [24]. **The same three causes recur: information from the future, a narrow set of inputs, and scores that do not reflect whether an alarm would help.**

[15] Reyna et al., Crit Care Med 2020 [21] Morrill et al., Crit Care Med 2020 [22] Wong et al., JAMA Intern Med 2021 [23] Fleuren et al., Intensive Care Med 2020 [24] Moor et al., Front Med 2021

RESEARCH QUESTION

The question this research asks

Using only what a clinician could have known at each hour, can a model warn of sepsis hours before onset with positive clinical utility, and does that utility hold in a hospital system it has never seen?

What has mostly been asked

How high can AUROC go on one prepared dataset, with one team's sepsis definition [23, 24]?

What this project asks

Would the alarm have helped this patient, at a false-alarm burden clinicians accept, and would it still help in the next hospital?

RQ1 Predictors

Does the whole clinical picture known at that hour (treatments, devices, admission context, history) add signal beyond vital signs and laboratory values?

RQ2 Architecture

Once causality is enforced, does organ-structured modelling add anything over simpler models?

RQ3 Transfer and trust

Does positive utility survive in another hospital system, and would healthcare professionals act on the alert?

[23] Fleuren et al., Intensive Care Med 2020 [24] Moor et al., Front Med 2021

APPROACH

Healthcare professionals in the loop make clinical AI safer and more useful

Better together

AI support improved clinicians' diagnostic accuracy beyond AI or clinicians alone [25]

Engaged clinicians save lives

Sepsis alerts confirmed by clinicians within 3 h: 18.7 % lower relative mortality [4]

Trust is built, not assumed

Clinicians trust AI through experience, endorsement and systems that respect their judgement [26]

Clinical input at every stage

From problem definition to deployment [27]; health records encode care decisions [16]



Example 1: antibiotics

Usually: every antibiotic order is an input, so the model predicts sepsis from the treatment that defines it, or antibiotics are dropped entirely.

Here: cefazolin at induction of anaesthesia, stopped within 24 h, with no culture sent, is surgical prophylaxis [28]. Prophylaxis is kept as a predictor by guideline rule; empirical and culture-directed antibiotics are excluded [29].



Example 2: suspicion proxies

Usually: a stress-dose hydrocortisone order or a chest X-ray request is treated as a strong predictor.

Here: these show that the team already suspects sepsis. They are declared as proxies and every result is reported with and without them, separating the patient's signal from the team's.

[4] Adams et al., Nat Med 2022 [16] Agniel et al., BMJ 2018 [25] Tschandl et al., Nat Med 2020 [26] Henry et al., NPJ Digit Med 2022 [27] Wiens et al., Nat Med 2019 [28] Bratzler et al., Am J Health Syst Pharm 2013 [29] Evans et al., Intensive Care Med 2021

Three open intensive-care databases, all accessed through PhysioNet

MIMIC-IV 3.1

Beth Israel Deaconess Medical Center, Boston

about 94,000

ICU stays

Every charted vital sign, laboratory result, medication, procedure, microbiology result and diagnosis, de-identified [30]

Role here: Development and internal testing

eICU-CRD 2.0

208 US hospitals, 2014 to 2015

200,859

ICU admissions

Tele-ICU records from many hospitals with different charting habits; times stored as minutes from ICU admission [31]

Role here: Second hospital system: the transfer test

PhysioNet 2019

Challenge data from US hospitals

40,336

patients

40 hourly variables, prepared by the organisers, with a pre-shifted sepsis label [15]

Role here: Comparison with published challenge models

Access requires credentialing, human-subjects research training and a data use agreement [32]. No patient-level data leave the analysis environment. One sepsis definition is applied to both raw databases with published benchmark code [33, 34].

[15] Reyna et al., Crit Care Med 2020 [30] Johnson et al., Sci Data 2023 [31] Pollard et al., Sci Data 2018 [32] Goldberger et al., Circulation 2000 [33] van de Water et al., ICLR 2024 [34] Bennett et al., GigaScience 2023

The research pipeline, built like the models it trains

Phase 1 Quantitative

Raw records

MIMIC-IV and eICU

Why: two hospital systems make transfer real

→

Harmonise

same concept, unit and clock

Why: compare like with like

→

One label

Sepsis-3 onset hour [35]

Why: results comparable

→

Hourly features

655, each used only from the hour recorded

Why: no information from the future

→

Models

16 compared on identical hours

Why: architecture is a hypothesis

→

Evaluate

utility, false alarms per day, transfer, sealed holdout

Why: what would help patients

Phase 2 Qualitative (in progress)

Online interviews and round-table discussions with 20 infectious disease and critical care consultants from different hospitals: would they trust and act on the alert, what explanation do they need, what alarm burden is acceptable [26].

→

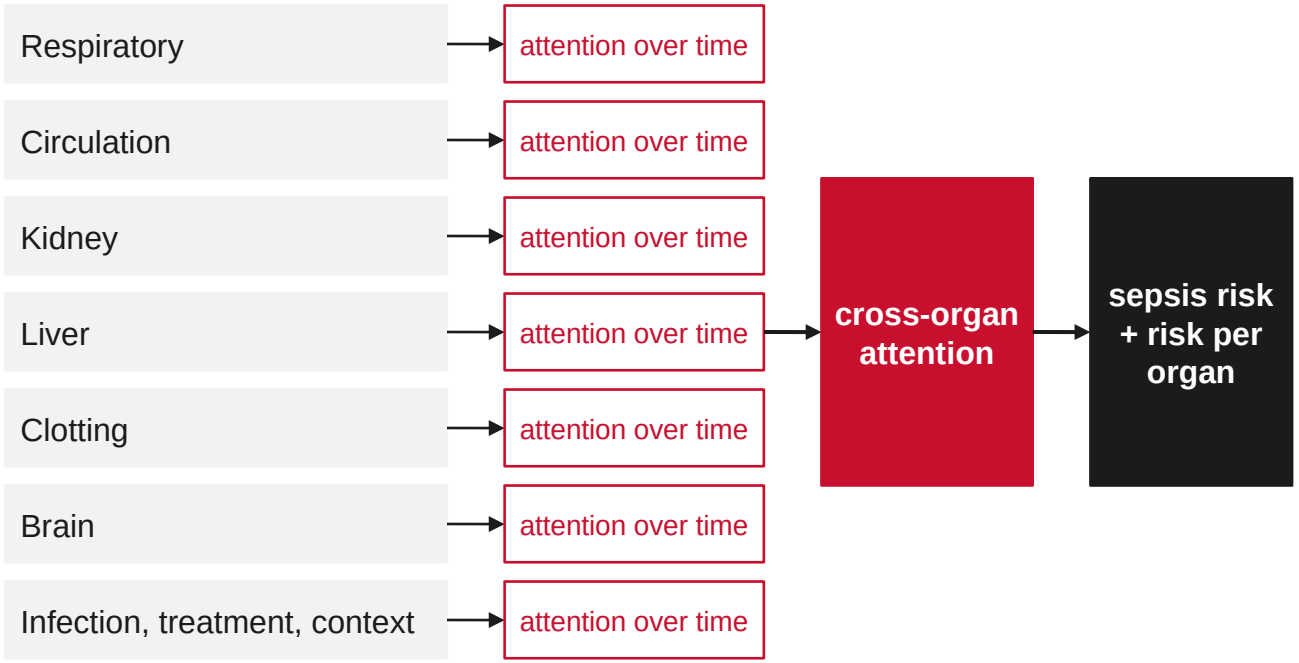
Phase 3 Future direction: hospital patients

A silent test on live ICU data (alerts logged, not shown, compared with outcomes), then a clinician-facing pilot [13].

[13] Kwong et al., Front Digit Health 2022 [26] Henry et al., NPJ Digit Med 2022 [35] Seymour et al., JAMA 2016

Organ-structured attention is a hypothesis we test, not an assumption

OSTA: organ-system-aware temporal attention (this project)



Mirrors how an ICU team reasons: each organ, then how organs affect each other. Built on transformer attention [36].

Compared on identical hours, rules fixed in advance

- **OSTA without organ grouping**
  the ablation that decides whether organ structure stays
- **CNN-Transformer (earlier work)**
  local patterns over 1 to 12 h, then attention
- **GRU-D, temporal CNN**
  models built for irregular ICU data [37, 38]
- **Signatures + boosted trees**
  the PhysioNet 2019 winner's method [21]
- **Self-supervised pretraining**
  learn from unlabelled stays first [39]
- **Logistic regression, boosted trees**
  simple baselines

Training loss chosen by clinical utility, against plain and time-smoothed losses [40].

[21] Morrill et al., Crit Care Med 2020 [36] Vaswani et al., NeurIPS 2017 [37] Che et al., Sci Rep 2018 [38] Bai et al., arXiv 2018 [39] Yeche et al., ICML 2021 [40] Yeche et al., ICML 2023

PRELIMINARY | full results in the manuscript in preparation

Result 1: making two hospital systems speak the same language

Example: body temperature

MIMIC-IV [30]

item 223761: Temperature, Fahrenheit

item 223762: Temperature, Celsius

clock: date-shifted date and time

eICU [31]

vital-sign table: temperature (often empty)

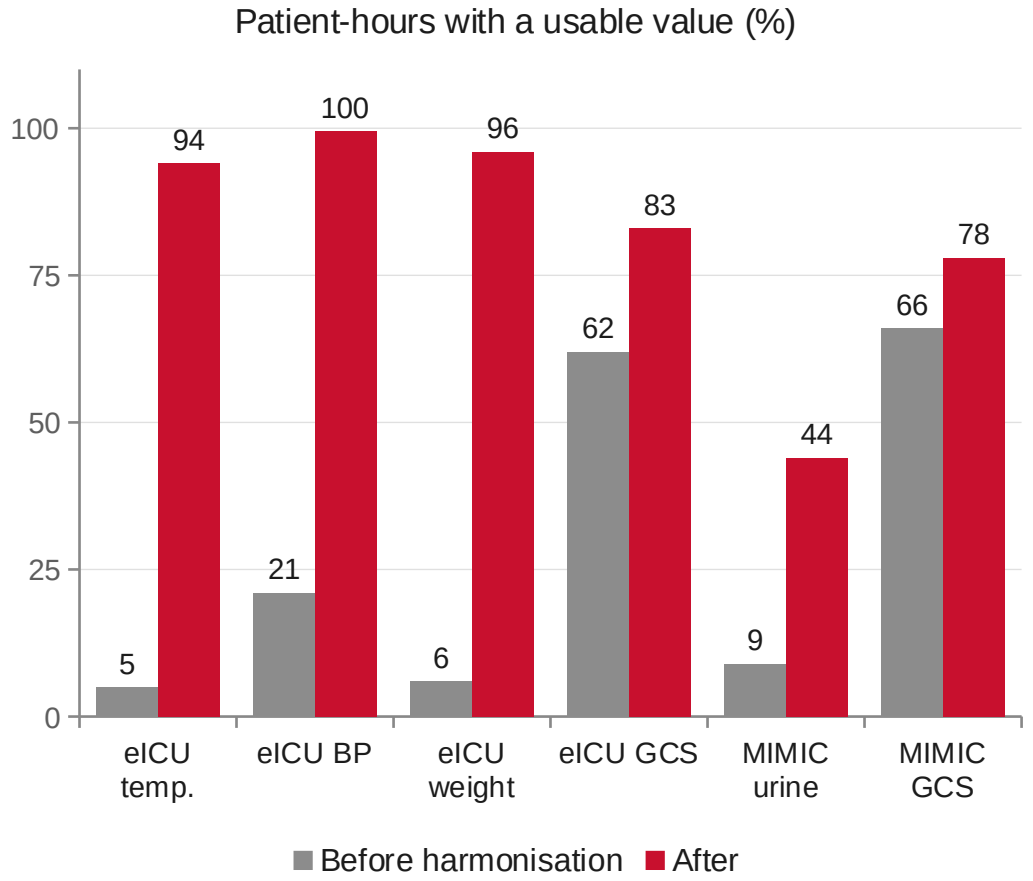
nurse charting: "Temperature (C)", "(F)"

clock: minutes from ICU admission

One concept

Temperature in °C, hours from ICU admission, usable from the hour it was charted

Repeated for 76 measurement channels plus drugs, devices, fluids and diagnoses [41].



[30] Johnson et al., Sci Data 2023 [31] Pollard et al., Sci Data 2018 [41] This work, preliminary, unpublished

PRELIMINARY | full results in the manuscript in preparation

Result 2: removing information from the future is the biggest single correction

An audit of our own earlier pipeline found

- Hours after sepsis onset labelled positive: the model learnt sepsis that had already started.
- Values used from when they were measured, not when they became available.
- Treatments that define the label among the inputs.
- Those models were retired and are not reported.

Leakage of this kind is documented across science [18].

Rebuilt causally: OSTA on the PhysioNet 2019 test set, 3-seed ensemble [15, 41]

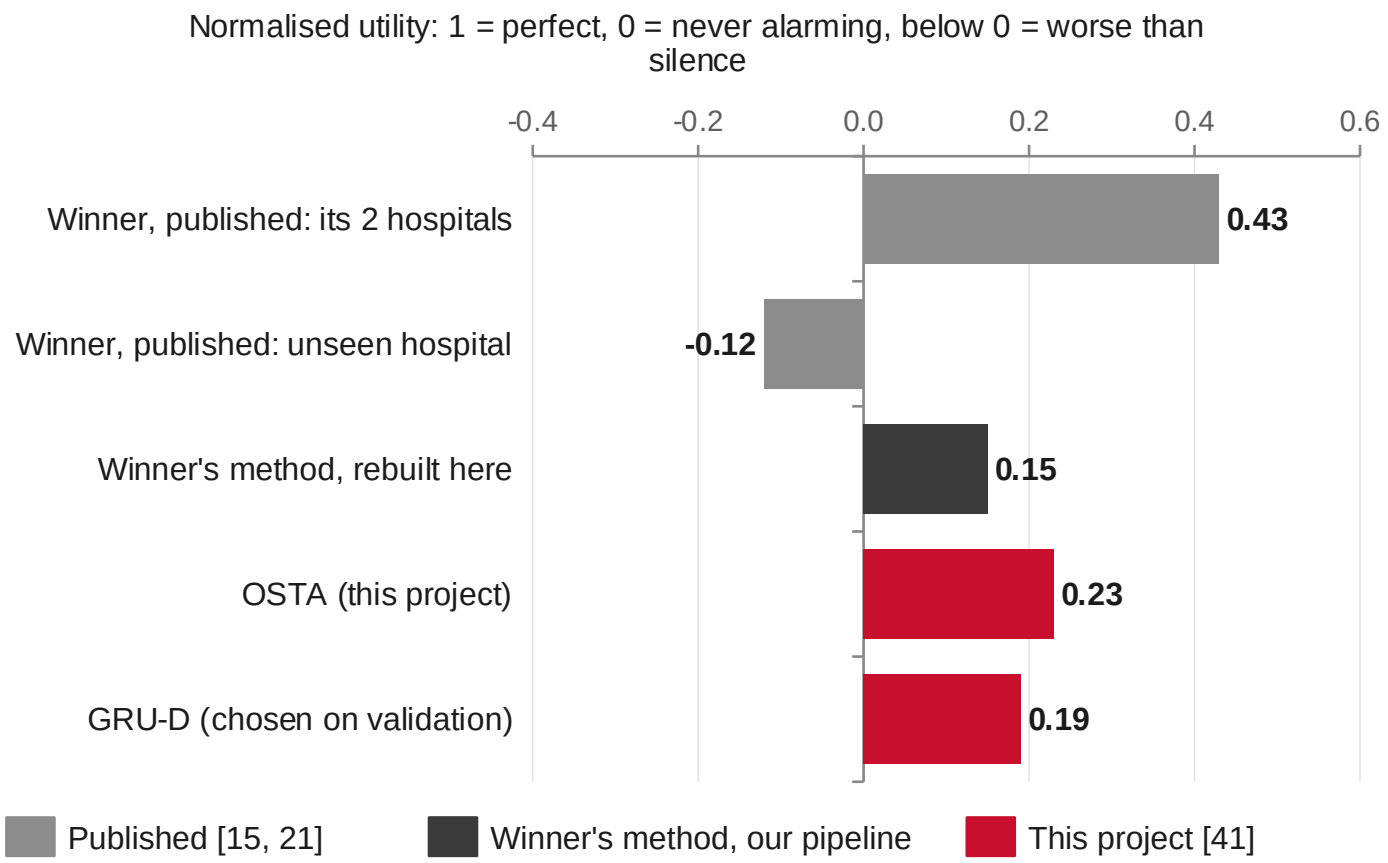
0.81	0.80	0.22
patient-level AUROC	hourly AUROC	utility (0.30 at the conformal threshold)

- Only 1.6 % of hours are positive, so ranking scores look modest even for a useful model.
- Utility 0.22: the alarms are worth about a fifth of perfect alarms after paying for false alarms.
- This honest baseline is where every later number starts.

[15] Reyna et al., Crit Care Med 2020 [18] Kapoor et al., Patterns 2023 [41] This work, preliminary, unpublished

PRELIMINARY | full results in the manuscript in preparation

Result 3: on identical MIMIC-IV hours, all 21 models have positive utility, including the winner's method



MIMIC-IV development test split

1,422 stays, 71 sepsis events, identical hours

21 / 21

models with utility above zero, every 95 % CI above zero

0.08-0.29

utility range across all models

0.23

OSTA, 95 % CI 0.13 to 0.34

0.15

winner's method, 95 % CI 0.06 to 0.22

Rules fixed in advance, applied on validation: organ grouping did not beat the same model without it; GRU-D, TCN and flat OSTA shortlisted. Validation and test rankings disagree, so the full cohort decides.

[15] Reyna et al., Crit Care Med 2020 [21] Morrill et al., Crit Care Med 2020 [41] This work, preliminary, unpublished

An honest, transferable test of sepsis warnings, and the path to hospital patients

Contribution so far

Causal, cross-database pipeline

one sepsis definition, two hospital systems, 655 clinically defined predictors [41]

Evaluation by clinical utility

false alarms per day, warning time, transfer, sealed holdout [15]

Three findings

leakage inflates scores; data harmonisation drives transfer; small samples overfit

Three manuscripts submitted in 2026

AMR diagnostics review [5]; sepsis-onset cohort study [42]; thermography meta-analysis [43]

Next steps

Now	Complete the qualitative study (20 infectious disease and critical care consultants); finish the 16-model comparison
Next	Full cohort (about 94,000 + 200,000 stays), three seeds, leave-hospital-out testing
Then	Freeze the protocol; open the sealed holdout once; set the alarm policy with the qualitative findings
Future	Silent test on hospital patients, then a clinician-facing pilot [13]

Numbers shown today are preliminary development-split results; the full results are in the manuscript in preparation.

[5] Elrggal et al., submitted 2026 [13] Kwong et al., Front Digit Health 2022 [15] Reyna et al., Crit Care Med 2020 [41] This work, preliminary, unpublished [42] Elrggal et al., submitted 2026 [43] Elrggal et al., submitted 2026

Where I would like your input

Which score should choose the model when they disagree?

Utility rewards an alarm hours before onset and charges for every false-alarm hour; AUROC rewards ranking [15, 19]. On our data they already rank models differently. How much sensitivity would you trade to halve false alarms per patient-day?

How should one sepsis label work across two hospital systems?

In MIMIC-IV, suspicion of infection needs an antibiotic and a culture; eICU records antibiotics only [35, 31]. Harmonise both to the weaker definition, keep each database's best definition, or treat them as two targets?

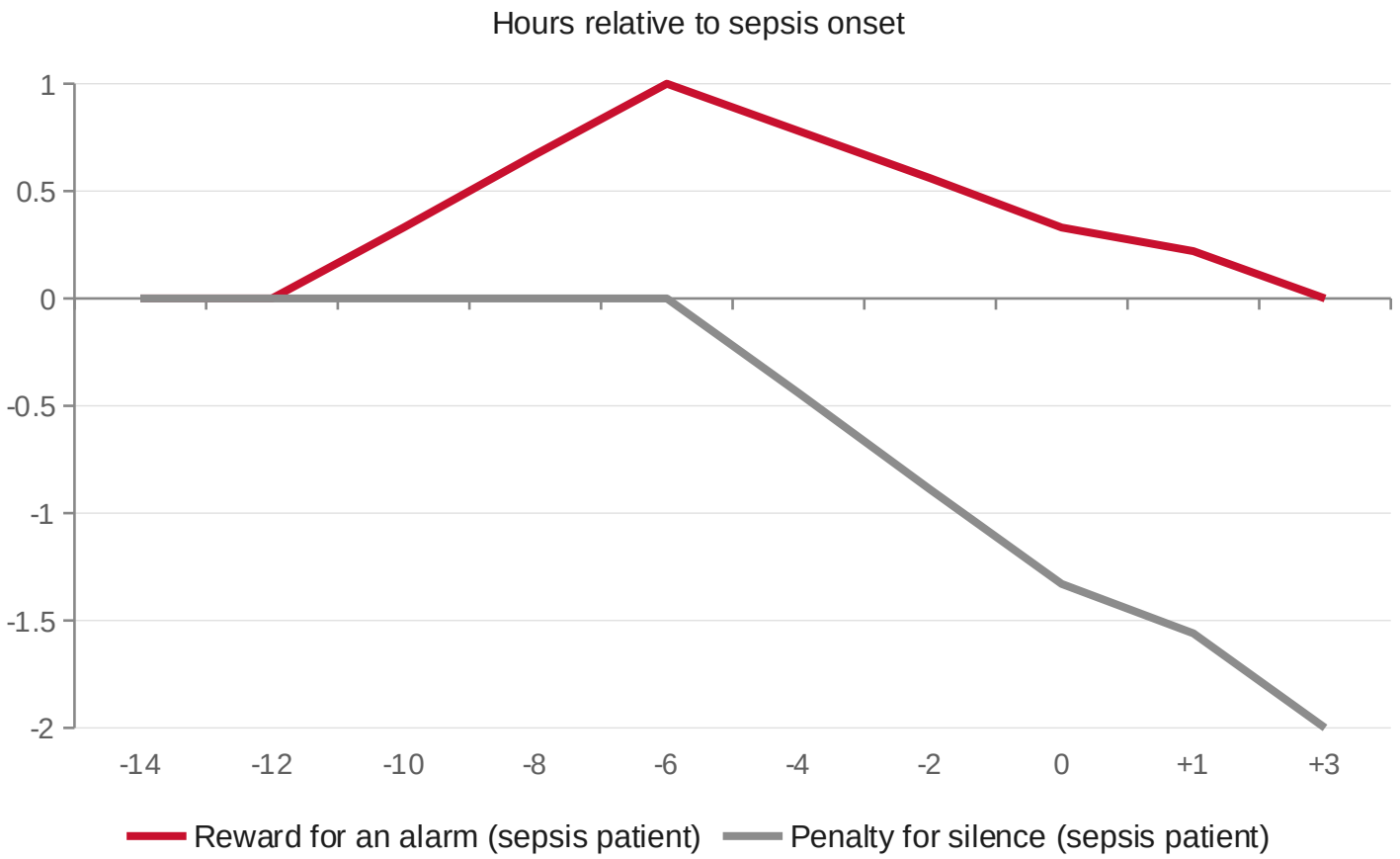
[15] Reyna et al., Crit Care Med 2020 [19] Shah et al., JAMA 2019 [31] Pollard et al., Sci Data 2018 [35] Seymour et al., JAMA 2016

Glossary of clinical terms

Sepsis-3	Organ dysfunction caused by a dysregulated response to infection; operationalised as suspected infection plus a SOFA rise of 2 or more [6]
Suspected infection	An antibiotic and a culture within 72 h / 24 h of each other [35]; eICU has antibiotics only
qSOFA, NEWS2	Bedside screening scores from vital signs and mental state [35, 44]; predictors here, never the label
Utility	Challenge score: reward for alarms up to 12 h before onset, cost for false alarms and missed sepsis; 1 perfect, 0 never alarming [15]
SOFA	0 to 4 points for each of six organ systems [7]
Onset and horizon	Onset is the first hour both criteria are met; the model asks every hour whether onset falls within the next 6 h
AKI stage	Kidney injury stage from creatinine and urine output [45]
Causal	Each value used only from the hour it was recorded, with a measured flag and its age; no hour after onset used

[6] Singer et al., JAMA 2016 [7] Vincent et al., Intensive Care Med 1996 [15] Reyna et al., Crit Care Med 2020 [35] Seymour et al., JAMA 2016 [44] Royal College of Physicians 2017
[45] KDIGO 2012

The utility score: what an alarm is worth, hour by hour [15]



- An alarm earns most at 6 h before onset; earlier than 12 h earns nothing.
- Each false-alarm hour on a patient without sepsis costs 0.05.
- Silence after onset costs up to 2 points per hour.
- Normalised: (score - never alarming) / (perfect - never alarming).
- Official scorer code used unchanged.

[15] Reyna et al., Crit Care Med 2020

Six choices that differ from common practice

Common practice	This project	Why
Challenge-prepared tables [15]	**Raw MIMIC-IV and eICU**	Transfer across hospital systems is tested for real
Each study its own definition [24]	**One Sepsis-3 definition on both databases [33, 34]**	Results comparable
Values carried forward; post-onset hours positive	**Each value used from the hour recorded, expires after 24 h; post-onset hours excluded**	No information from the future [18]
40 to 125 variables	**655 clinically defined predictors**	The whole clinical picture
Antibiotics as predictors	**Label-defining antibiotics excluded; prophylaxis kept by rule [28]**	Avoids predicting the label from itself
AUROC on one split	**Utility, false alarms per day, warning time, calibration, sealed holdout**	What would help patients [19]

[15] Reyna et al., Crit Care Med 2020 [18] Kapoor et al., Patterns 2023 [19] Shah et al., JAMA 2019 [24] Moor et al., Front Med 2021 [28] Bratzler et al., Am J Health Syst Pharm 2013
[33] van de Water et al., ICLR 2024 [34] Bennett et al., GigaScience 2023

The 655 predictors, each with a measured flag and an age

Vital signs and monitoring

heart rate, cuff and arterial pressures, respiratory rate, oxygen saturation, temperature, end-tidal CO2, CVP, cardiac output, pain and sedation scores

Laboratory and blood gas

chemistry, full blood count with differential, coagulation, lactate, procalcitonin, troponin, BNP, ferritin, D-dimer, ammonia, urinalysis, blood gases

Scores and derived

SOFA per organ [7], qSOFA, NEWS2 [44], shock index, P/F and S/F ratios, changes from previous and baseline, 24 h extremes

Organ support

vasopressor dose (noradrenaline equivalents), ventilation settings, oxygen flow, renal replacement, urine output and AKI stage [45]

Fluids, drugs, devices

crystalloids, transfusions, nutrition, central and arterial lines, catheters, drains; steroids, immunosuppressants, chemotherapy, sedatives, opioids

Context and history

admission source, ICU type, hours before ICU, prior admissions, prior positive cultures and antibiotics, 17 comorbidities, imaging orders

Prophylactic antibiotics

surgical prophylaxis by agent, timing, duration, no culture or escalation [28]; medical prophylaxis by agent and dose

Never predictors

label-defining antibiotics and cultures; race, ethnicity, insurance (fairness analysis only); anything after the prediction hour

[7] Vincent et al., Intensive Care Med 1996 [28] Bratzler et al., Am J Health Syst Pharm 2013 [44] Royal College of Physicians 2017 [45] KDIGO 2012

Every model compared (1 of 2): baselines, the winner's method and sequence models

Model	Core idea	How it differs	Utility
Logistic regression	One weighted sum of the current hour's predictors	No time modelling: the floor every model must beat	**0.08**
Boosted trees [46]	Many small decision trees, each correcting the last	Non-linear thresholds, but sees one hour at a time	**0.10**
Signatures + boosted trees [21]	Recent trajectories summarised as path signatures, then boosted trees	The PhysioNet 2019 winner's method; time enters through fixed summaries	**0.15**
OSTA (earlier work)	Predictors grouped by organ, attention over time per organ, then across organs	The reference model; risk per organ	**0.23**
Output fusion (earlier work)	OSTA plus small per-organ models trained on separate patients, added at the output	Independent per-organ second opinions fused at the end	**0.25**
CNN-Transformer (earlier work) [36]	Convolution finds 1 to 12 h patterns, then attention	No organ grouping	**0.24**
Flat OSTA	OSTA with every predictor in one group	Removes only the organ grouping: the decisive test	**0.23**
GRU-D [37]	Recurrent memory; values fade towards normal as they age	Models missing and stale data explicitly	**0.19**
TCN [38]	Stacked causal convolutions with a widening reach	No attention; fixed look-back	**0.17**

Utility: normalised utility on the MIMIC-IV development test split, one seed, preliminary [41]. Every 95 % CI is above zero and the intervals overlap, so the order is not a ranking.

[21] Morrill et al., Crit Care Med 2020 [36] Vaswani et al., NeurIPS 2017 [37] Che et al., Sci Rep 2018 [38] Bai et al., arXiv 2018 [46] Friedman, Ann Stat 2001

Every model compared (2 of 2): channel encoders, self-supervised pretraining, ensemble

Model	Core idea	How it differs	Utility
CNN to OSTA	A small convolution per channel over value, measured flag and age, then OSTA	Learns each signal's own pattern before organ grouping	**0.28**
Event OSTA [47]	Each hour attends to fresh vital-sign measurements of the last 24 h	Uses new measurements only, not carried-forward values	**0.21**
Self-supervised CNN to OSTA [39]	The same encoder first trained on unlabelled stays to forecast values and organ scores	Learns physiology before it sees a sepsis label	**0.29**
Self-supervised CNN, flat head	The same pretrained encoder with one group	Its organ ablation; added as a declared exploratory pair	**0.21**
Pooled encoder twins	Encoder pretrained on both databases	Adaptation across hospital systems	**0.23 / 0.20**
Hazard head [48]	Risk at 1, 3, 6, 12 and 24 h as a survival curve	Uses when onset happens, not only whether	**0.18**
Mean ensemble	Average of OSTA, CNN-Transformer and self-supervised CNN to OSTA	No learned weights	**0.28**

Retired in earlier versions: gated and ungated attention and observer features, which did not differ from OSTA. Utility: normalised utility on the MIMIC-IV development test split, one seed, preliminary [41]. Every 95 % CI is above zero and the intervals overlap, so the order is not a ranking.

[39] Yeche et al., ICML 2021 [47] Horn et al., ICML 2020 [48] Gensheimer and Narasimhan, PeerJ 2019

Cohorts, splits and the sealed holdout

Database	Candidate stays	Sample so far	After exclusions	Sealed holdout
MIMIC-IV 3.1 [30]	about 94,000	10,000	8,795	20 % of patients, deterministic hash
eICU-CRD 2.0 [31]	200,859	10,000	5,710	20 % of patients, deterministic hash

- Labels: Sepsis-3 via YAIB / ricu [33, 34]; onset within the next 6 h; post-onset hours excluded; up to 120 h per stay.
- The 10,000-stay samples are fixed random draws, identical in every run.
- The holdout is opened once, after the protocol is frozen; nothing is tuned on it.
- Unlabelled pretraining corpus: 67,455 MIMIC-IV and 75,000 eICU stays outside the cohort and the holdout.

[30] Johnson et al., Sci Data 2023 [31] Pollard et al., Sci Data 2018 [33] van de Water et al., ICLR 2024 [34] Bennett et al., GigaScience 2023

BACKUP

On 10,000 stays models overfit fast and the validation split is too small to rank them

1 to 3

epochs to the best validation score, while training loss keeps falling

175

septic stays per training split, against 300 to 655 predictors

0.12 vs 0.28

validation utility of OSTA and flat OSTA, which tie on test (0.23): validation and test rankings disagree

- Response: full cohort (about nine times more sepsis cases), self-supervised pretraining on unlabelled stays [39], loss chosen by utility [40], three seeds.
- All figures: this work, preliminary [41].

[39] Yeche et al., ICML 2021 [40] Yeche et al., ICML 2023 [41] This work, preliminary, unpublished

RD6 supplement: publications and outputs

Manuscript	Journal	Submitted	Status
Artificial Intelligence in the Diagnosis of Antimicrobial Resistance: Rapid Diagnostics and Imaging Innovations	Int J Antimicrob Agents	9 Oct 2026	Technical check
Temporal Patterns and Predictors of Sepsis Onset in ICU Patients: A Retrospective Cohort Study Using PhysioNet Data	Frontiers in Medicine	23 Sep 2026	Editorial assignment
Diagnostic Applications of Infrared Thermography Across Infectious Diseases: A Systematic Review with a Subgroup Meta-Analysis	JMIR	18 Jul 2026	In review
Early sepsis warning on MIMIC-IV and eICU: causal predictors, utility and transfer	critical care informatics journal	in preparation	full-cohort and holdout results pending

Earlier CNN-Transformer and OSTA manuscripts were withheld after the leakage audit; both architectures are re-evaluated causally in the current comparison.

RD6 supplement: training and the March 2026 plan

Training

- Harvard PPCR: Clinical Trials Design, Implementation and Analysis (completed Feb 2026)
- Iowa State University: Machine Learning Programme (completed Dec 2025)
- DataCamp Python Data Science Career Track (in progress)
- PhysioNet credentialed access: MIMIC-IV 3.1, eICU-CRD 2.0

March 2026 plan: what happened

- External validation on MIMIC-IV: became the core programme, extended to eICU.
- CNN-Transformer and OSTA papers: withheld after the leakage audit.
- AMR review: revised and submitted to Int J Antimicrob Agents.
- Clinician study: now the qualitative phase (20 consultants), nearly complete.
- Conference presentation: this talk.

References (1 of 2)

[1] Rudd KE, Johnson SC, Agesa KM, et al. Global, regional, and national sepsis incidence and mortality, 1990-2017: analysis for the Global Burden of Disease Study. Lancet. 2020;395:200-211.

[2] Antimicrobial Resistance Collaborators. Global burden of bacterial antimicrobial resistance in 2019: a systematic analysis. Lancet. 2022;399:629-655.

[3] Timbrook TT, Morton JB, McConeghy KW, et al. The effect of molecular rapid diagnostic testing on clinical outcomes in bloodstream infections: a systematic review and meta-analysis. Clin Infect Dis. 2017;64:15-23.

[4] Adams R, Henry KE, Sridharan A, et al. Prospective, multi-site study of patient outcomes after implementation of the TREWS machine learning-based early warning system for sepsis. Nat Med. 2022;28:1455-1460.

[5] Elrggal ME, et al. Artificial intelligence in the diagnosis of antimicrobial resistance: rapid diagnostics and imaging innovations. Submitted, Int J Antimicrob Agents, 2026.

[6] Singer M, Deutschman CS, Seymour CW, et al. The Third International Consensus Definitions for Sepsis and Septic Shock (Sepsis-3). JAMA. 2016;315:801-810.

[7] Vincent JL, Moreno R, Takala J, et al. The SOFA (Sepsis-related Organ Failure Assessment) score to describe organ dysfunction/failure. Intensive Care Med. 1996;22:707-710.

[8] Liu V, Escobar GJ, Greene JD, et al. Hospital deaths in patients with sepsis from 2 independent cohorts. JAMA. 2014;312:90-92.

[9] Seymour CW, Gesten F, Prescott HC, et al. Time to treatment and mortality during mandated emergency care for sepsis. N Engl J Med. 2017;376:2235-2244.

[10] Kumar A, Roberts D, Wood KE, et al. Duration of hypotension before initiation of effective antimicrobial therapy is the critical determinant of survival in human septic shock. Crit Care Med. 2006;34:1589-1596.

[11] Churpek MM, Snyder A, Han X, et al. Quick SOFA, SIRS, and early warning scores for detecting clinical deterioration in infected patients outside the ICU. Am J Respir Crit Care Med. 2017;195:906-911.

[12] Yu AC, Mohajer B, Eng J. External validation of deep learning algorithms for radiologic diagnosis: a systematic review. Radiol Artif Intell. 2022;4:e210064.

[13] Kwong JCC, Erdman L, Khondker A, et al. The silent trial: the bridge between bench-to-bedside clinical AI applications. Front Digit Health. 2022;4:929508.

[14] Nagendran M, Chen Y, Lovejoy CA, et al. Artificial intelligence versus clinicians: systematic review of design, reporting standards, and claims of deep learning studies. BMJ. 2020;368:m689.

[15] Reyna MA, Josef CS, Jeter R, et al. Early prediction of sepsis from clinical data: the PhysioNet/Computing in Cardiology Challenge 2019. Crit Care Med. 2020;48:210-217.

[16] Agniel D, Kohane IS, Weber GM. Biases in electronic health record data due to processes within the healthcare system: retrospective observational study. BMJ. 2018;361:k1479.

[17] Zech JR, Badgeley MA, Liu M, et al. Variable generalization performance of a deep learning model to detect pneumonia in chest radiographs: a cross-sectional study. PLoS Med. 2018;15:e1002683.

[18] Kapoor S, Narayanan A. Leakage and the reproducibility crisis in machine-learning-based science. Patterns. 2023;4:100804.

[19] Shah NH, Milstein A, Bagley SC. Making machine learning models clinically useful. JAMA. 2019;322:1351-1352.

[20] Kelly CJ, Karthikesalingam A, Suleyman M, Corrado G, King D. Key challenges for delivering clinical impact with artificial intelligence. BMC Med. 2019;17:195.

[21] Morrill JH, Kormilitzin A, Nevado-Holgado AJ, et al. Utilization of the signature method to identify the early onset of sepsis from multivariate physiological time series in critical care monitoring. Crit Care Med. 2020;48:e976-e981.

[22] Wong A, Otles E, Donnelly JP, et al. External validation of a widely implemented proprietary sepsis prediction model in hospitalized patients. JAMA Intern Med. 2021;181:1065-1070.

[23] Fleuren LM, Klausch TLT, Zwager CL, et al. Machine learning for the prediction of sepsis: a systematic review and meta-analysis of diagnostic test accuracy. Intensive Care Med. 2020;46:383-400.

[24] Moor M, Rieck B, Horn M, Jutzeler CR, Borgwardt K. Early prediction of sepsis in the ICU using machine learning: a systematic review. Front Med. 2021;8:607952.

References (2 of 2)

[25] Tschandl P, Rinner C, Apalla Z, et al. Human-computer collaboration for skin cancer recognition. Nat Med. 2020;26:1229-1234.

[26] Henry KE, Kornfield R, Sridharan A, et al. Human-machine teaming is key to AI adoption: clinicians' experiences with a deployed machine learning system. NPJ Digit Med. 2022;5:97.

[27] Wiens J, Saria S, Sendak M, et al. Do no harm: a roadmap for responsible machine learning for health care. Nat Med. 2019;25:1337-1340.

[28] Bratzler DW, Dellinger EP, Olsen KM, et al. Clinical practice guidelines for antimicrobial prophylaxis in surgery. Am J Health Syst Pharm. 2013;70:195-283.

[29] Evans L, Rhodes A, Alhazzani W, et al. Surviving Sepsis Campaign: international guidelines for management of sepsis and septic shock 2021. Intensive Care Med. 2021;47:1181-1247.

[30] Johnson AEW, Bulgarelli L, Shen L, et al. MIMIC-IV, a freely accessible electronic health record dataset. Sci Data. 2023;10:1.

[31] Pollard TJ, Johnson AEW, Raffa JD, et al. The eICU Collaborative Research Database, a freely available multi-center database for critical care research. Sci Data. 2018;5:180178.

[32] Goldberger AL, Amaral LAN, Glass L, et al. PhysioBank, PhysioToolkit, and PhysioNet: components of a new research resource for complex physiologic signals. Circulation. 2000;101:e215-e220.

[33] van de Water R, Schmidt HNA, Elbers PWG, et al. Yet Another ICU Benchmark: a flexible multi-center framework for clinical machine learning. ICLR 2024.

[34] Bennett N, Plecko D, Ukor IF, Meinshausen N, Buhlmann P. ricu: R's interface to intensive care data. GigaScience. 2023;12.

[35] Seymour CW, Liu VX, Iwashyna TJ, et al. Assessment of clinical criteria for sepsis (Sepsis-3). JAMA. 2016;315:762-774.

[36] Vaswani A, Shazeer N, Parmar N, et al. Attention is all you need. Advances in Neural Information Processing Systems (NeurIPS). 2017.

[37] Che Z, Purushotham S, Cho K, Sontag D, Liu Y. Recurrent neural networks for multivariate time series with missing values. Sci Rep. 2018;8:6085.

[38] Bai S, Kolter JZ, Koltun V. An empirical evaluation of generic convolutional and recurrent networks for sequence modeling. arXiv:1803.01271. 2018.

[39] Yeche H, Dresdner G, Locatello F, Huser M, Ratsch G. Neighborhood contrastive learning applied to online patient monitoring. International Conference on Machine Learning (ICML). 2021.

[40] Yeche H, Pace A, Ratsch G, Kuznetsova R. Temporal label smoothing for early event prediction. International Conference on Machine Learning (ICML). 2023.

[41] Elrggal ME, et al. Early sepsis warning on MIMIC-IV and eICU: preliminary development-split results, October 2026. Unpublished; manuscript in preparation.

[42] Elrggal ME, et al. Temporal patterns and predictors of sepsis onset in ICU patients: a retrospective cohort study using PhysioNet data. Submitted, Front Med, 2026.

[43] Elrggal ME, et al. Diagnostic applications of infrared thermography across infectious diseases: a systematic review with a subgroup meta-analysis. Submitted, JMIR, 2026.

[44] Royal College of Physicians. National Early Warning Score (NEWS) 2: standardising the assessment of acute-illness severity in the NHS. London: RCP; 2017.

[45] Kidney Disease: Improving Global Outcomes (KDIGO) AKI Work Group. KDIGO clinical practice guideline for acute kidney injury. Kidney Int Suppl. 2012;2:1-138.

[46] Friedman JH. Greedy function approximation: a gradient boosting machine. Ann Stat. 2001;29:1189-1232.

[47] Horn M, Moor M, Bock C, Rieck B, Borgwardt K. Set functions for time series. International Conference on Machine Learning (ICML). 2020.

[48] Gensheimer MF, Narasimhan B. A scalable discrete-time survival model for neural networks. PeerJ. 2019;7:e6257.